

[L3] Chapters 5 & 6 - The sound of software

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Until the early 1960s, most jobs were being computed by human computers, even though automatic computers were there. They were unreliable and didn't work most of the time. The times that they work, they were used.

Part 2: Computing Sounds

When EDSAC was built, Maurice Wilkes in Cambridge was very proud of it. To see if the computer or the program was working, you could listen to the different components (ex: vacuum tubes). There were no screens to display the progress of the program. That was done by listening in to the software.

If you understood how the computers worked internally, you could make them humm music like an instrument.

Later machines that were built with transistors did not make sound. Therefore, engineers put in speakers to be able listen in.

The UNIVAC had a "Stall Speaker" button. This means that by default, the speaker was on. The standard use case was with the speaker on, so that they would know what was happening inside the computer.

This is the case for most early computers, because there was no other visual signal.

Sounds were used for various reasons:

- Announcing the program was terminated
- Auditive monitoring (it works)
- Specific navigation within a program
- Part of the instruction (Knowing where in the program the computer currently is)

Soon, computer sounds started becoming parts of games, software. Music was being created on computers.

Part 3: Use of metaphor

To program meant to prepare a program

Programming started being used as a verb in 1948

Metaphors means appropriating a machine. Taking a machine and making it do something, sometimes that it's not meant to do, just to see if it works.

In the early 1950s, a program was not code. It was the wiring on a plugboard, a mess of cables that create the desired effect. Because of this, programming was also sometimes called plugging, pseudo coding, assembling, autocoding.

In the 1960s, programming languages became a metaphor. It was a language and the computer could **translate** it into wiring. They were taking new things and linking them to something familiar

Memory was another example. The mercury delay line could only store a bit for a very short time. Long term memory was, however, not necessary for the time's applications. They only needed to store numbers during calculations.

In the administration tradition, storing information for a long time was useful in the administration tradition, so they used magnetic tape, punched tape or punched cards in order to store information and potentially feed it back into the machine if needed.

In his 1984 book "Defining Technology", Bolter stated that, once a technology becomes ubiquitous

enough, the words used to describe it start being used to describe everyday life. The metaphor basically turns around (Education is "programming a child", taking a holiday is "wiping your harddrive")

Part 4: IBM and the European Dwarves

Within Europe there was an automation manufacturing industry as well before WW2. We are not talking about computers here yet, but various automation machines.

Odhner

HQ: St. Petersburg (1873)

It had a wide range of efficient and well respected machines being sold all over Europe. St. Petersburg was not really the place to be during the Russian Revolution though. However, Odhner clones were sold all over Europe after that.

Atvidaberg Industrie

HQ Sweden

Founded 1920

Produced very successful calculating machines and eventually changed their name to the name of the machine: Facit

1970 they went bankrupt and was sold to Electrolux

In Europe, they functioned on par with IBM, but they failed to innovate after WW2 and stuck to one kind of machines, which led to their bankruptcy.

Zuse KG

HQ Germany

Founded by Konrad Zuse

Was producing partly mechanical computing machines. They were very practical for computing purposes. Zuse was quite successful in obtaining contracts from the German government, who wanted to support an all-German computer. In 1967 he had to sell his company to Siemens AG.

Ferranti

HQ Great Britain

It was the company that commercialised the Manchester Baby. The company was fairly successful and remained in business into the 1980s

Bull

HQ France

In the 1930s, it sold office machinery in France. After the war it went into electronic computers (Gamma 3, Gamma 60). These were electronic computing machines meant for serving in administration and serving in automatic business. The French government bought Bull products and also encouraging all French companies to buy from Bull.

1962 it became too difficult to compete with IBM and General Electric stepped in by supplying capital.

1968 Honeywell took over Bull. Honeywell Bull became quite successful afterwards within Europe.

The fact that the French called the computer an "ordinateur" (a machine that orders) shows that they were approaching it more from an administrative perspective, whereas the American and other European countries' term "computer" suggests more of a scientific direction.

Electrologica

HQ The Netherlands

A design of the ARMAC that was put into commercial use (1956), called the Electrologica. This was produced by the Mathematical Centre

It was produced as **the** Dutch machine.

The Electrologica X1 was in competition with IBM's sales force. It functioned great, it was directed towards the scientific field and had support from dutch companies which liked the concept of an all-dutch machine.

Electrologica was successful in the field of science but they failed to move into the Administration business.

The Electrologica X8 fixed that. It had good features but the R&D was expensive and it sold very few units. 1966, it was bought by Philips.

Regnecentralen

HQ Denmark

Founded 1955

Had a successful range of computers that were also sold with stocks. This encouraged clients to keep buying Regnecentral to increase the value of their stocks. This model kept them in business into the 1980s. They also provided software adapted to the client's needs.

1989 they were taken over by International Computers Limited (a british company)

International Computers Limited

HQ Great Britain

Founded 1968

It was a merger of a number of smaller computer manufacturers (ex: Ferranti, Elliot, etc)

They were successful in selling IBM-compatible systems.

They are still in business under the name **Fujitsu**, who bought them in 2002

Eastern Europe:

- In Czechoslovakia, Antonin Svoboda, who had worked at MIT in the US, returned to Prague after the war and set out to form a laboratory for information processing.
- He had visited the UK and the US
- Together with mathematician Vaclav Hruska he started building a machine
- The country was communist. Therefore, their project was a state-organized company that would build machines for government purposes. No machines were sold to companies unless they were state-owned or could convince the government that they were a necessity.
- 1951 they designed the SAPO (similar to the ARRA, it could perform huge calculations)
- 1957 the SAPO was produced but was never a great success. Both Hruska and Svoboda had already moved on to another project
- EPOS (1) was realised in 1959. It was built from transistors and were an improvement upon the SAPO
- Because Svoboda had been to the US and UK, the government didn't trust him and his progress was slowed down.
- The Czechoslovakia joined the COMECON (the association of communist countries) quite late
- The United Systems of Electronic Computers was producing the EC(Electronic Computer) IBM clones for all the COMECON countries
- In Czechoslovakia, Zdenek Trnka was building analog machines to aid in steel productions

IBM and the European dwarves: a summary

- Until the 1980s, european companies were actually playing a role
- They were not too late. They all started about the time that UNIVAC started.
- These companies were funded by european companies
- What worked against these companies, was that they were interested in building scientific machines. They did not notice that administration was a huge market, where the big money was.

The majority realised too late that they would have to redirect resources there. In that market, they had to compete with IBM.

- Large companies like Philips and Siemens, who would have had the resources to compete with IBM, had agreements with IBM to not compete.
 - In the field of administration, they were sadly too little, too late
 - When governments had to retreat their funding after 1965, few companies had the resources to stay afloat.
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- 1960: 60% of machines in europe were IBM. 90% of machines in europe were not european
 - IBM's sales strategy was superior
 - With the arrival of the System/360, European companies were forced to build IBM compatibles or disappear